Part 1 — Quick Check

1. What is an "agent," in the sense used in this session?
    - A) A chatbot with a friendlier interface
    - B) **An LLM wrapped in a loop that can observe results and decide what to do next** ✓
    - C) A model trained on more recent data than usual
    - D) A human reviewing an LLM's answers before they're sent
2. Put the loop steps in the correct order.
    - A) Act → Think → Observe
    - B) Think → Act → Observe
    - C) **Observe → Think → Act** ✓
    - D) Observe → Act → Think
3. In the OBSERVE step, what is the agent actually looking at?
    - A) Its own training data
    - B) **The current state, or the result of its last action** ✓
    - C) A list of available tools only
    - D) The user's entire chat history across every past session
4. Why doesn't "more steps" automatically mean "more reliable"?
    - A) More steps always cost more money, which lowers reliability
    - B) **Each extra step is another place an error can be introduced, and mistakes compound across steps** ✓
    - C) Agents can only take a maximum of 3 steps before failing
    - D) More steps means the model has to use more of its context window
5. What actually changes about the underlying LLM itself when it's used inside an agent loop, versus a single plain call?
    - A) It's fine-tuned specifically for that task
    - B) It's given a larger context window automatically
    - C) **Nothing — it's the same model, called repeatedly, wrapped in code that decides when to call it again** ✓

- D) It's swapped for a smaller, faster model for intermediate steps

Part 2 — Short Answer

1. Describe what happens in each of the three loop steps:

In the **Observe** step, the agent examines the current state—either initial context or the result of the previous action. This is the ground truth it works from. In the **Think** step, the agent decides what to do next based on what it observed: it reasons about whether it has enough information, what tools might help, or if it's ready to answer. In the **Act** step, the agent executes that decision—calling a tool, searching for information, or providing a final answer. The loop then restarts, and the agent observes the outcome of its action.

2. What was the agent observing between searches in the live demo?

The agent searched once, then observed the results. Based on what it found, it recognized the information was incomplete or didn't directly answer the question, so it decided to search again with a different query or refined approach. The second search's results then gave it enough context to provide a confident answer. The key is that the agent wasn't just following a fixed script—it *adapted* based on what it learned from the first search.

3. What would you say to your teammate about looping 10 times instead of 3?

More loops don't guarantee better answers—each step is a chance for the agent to go off track or misinterpret results. If the agent makes a mistake early (wrong tool choice, misreading data), more loops just compound the problem. Better to design thoughtful loops with clear success criteria: *Does the agent have the info it needs? Is it confident? Has it checked for errors?* Three well-designed steps beat ten unfocused ones.

Part 3 — Applied Scenario: License Renewal Agent

1. Tools the agent would need:

- **License database tool** — to check current license details and expiration date
- **Price lookup tool** — to fetch the current renewal price
- **Historical pricing tool** — to access last year's price for comparison
- **Renewal approval/execution tool** — to process the renewal if conditions are met
- **Alert/notification tool** — to flag issues or confirmations

2. Observe → Think → Act cycles:

Cycle 1:

- **Observe:** Today's date and license expiration date retrieved from the database
- **Think:** Is renewal needed soon? (e.g., within 30 days?) If not, stop. If yes, proceed.
- **Act:** Fetch current renewal price

Cycle 2:

- **Observe:** Current price and last year's price are now available
- **Think:** Did the price increase by more than 10%? Calculate: (new - old) / old × 100
- **Act:** If ≤10% increase, approve renewal; if >10%, alert the team to decide

Cycle 3:

- **Observe:** Renewal approval status
- **Think:** Is the renewal approved? Can we proceed with payment?
- **Act:** Execute renewal transaction and log the transaction

Cycle 4 (optional):

- **Observe:** Renewal confirmation (success or failure)
- **Think:** Did the renewal complete? Is the license extended?
- **Act:** Send confirmation notification or escalate if failed

3. Silent failure point & mitigation:

Silent failure: The price lookup tool returns stale or incorrect data (e.g., a cached price from weeks ago, or an API timeout that returns a default value). The agent renews at what it thinks is within budget, but the actual charge is much higher.

How to catch it:

- Add a **data freshness check** — verify the price was retrieved within the last 24 hours
- Implement a **secondary validation** — fetch the price from two independent sources and flag if they differ by >5%
- Set a **price anomaly detector** — if the new price is 50%+ higher than the historical trend, require manual review before executing
- Log all price data with timestamps and sources for audit trail